

Multiphase equilibria for binary and ternary mixtures containing propionic acid, *n*-butanol, butyl propionate, and water

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Abstract

Isothermal vapor–liquid equilibrium (VLE) data were measured for propionic acid + butyl propionate at 373.15 and 393.15 K, and isothermal vapor–liquid–liquid equilibrium (VLLE) data were also measured for *n*-butanol + water, butyl propionate + water, and water + *n*-butanol + butyl propionate at temperatures ranging from 323.15 to 393.15 K. No azeotrope was found in propionic acid + butyl propionate. The mutual solubility data of the binary aqueous systems were correlated well with the NRTL model accompanying with temperature-dependent parameters. Improvement on the calculation of saturated vapor compositions has been made by using two-term virial equation with one adjustable binary interaction parameter to represent the non-ideality of the vapor phase. The model parameters determined from the binary VLLE data of *n*-butanol + water and butyl propionate + water and the binary VLE data of *n*-butanol + butyl propionate are capable of predicting satisfactorily the VLLE properties for the ternary system of water + *n*-butanol + butyl propionate.

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1. Introduction

Reactive distillation that combines reaction and distillation in a single unit has received a wide attention in the last decade [1]. This one-step process unit has several advantages over traditional two (or more)-step method, such as simplifying production flow sheet, improving productivity, minimizing waste production, and reducing energy requirement. It has been successfully applied in the manufacturing esters (e.g. methyl acetate) and ethers (e.g. methyl *tert*-butyl ether). Since chemical reaction and distillation implement simultaneously in a reactive distillation column, the performance of the column should depend on both the kinetics of reaction and the phase equilibrium behavior of mixtures. Multiphase coexistence may occur in the mixtures of water and esters, which are the products of esterification of acids with alcohols. Moreover, water is also partially miscible with *n*-butanol or heavier alkanols. The vapor–liquid equilibrium (VLE) and vapor–liquid–liquid equilibrium (VLLE) proper-

ties of mixtures containing water, esters, alcohols, and acids, therefore, are fundamentally important in the process development for production of esters via reactive distillation method.

The objective of this study is to investigate the multiphase coexistence behavior of aqueous mixtures containing butyl propionate and *n*-butanol, which may encounter in the production of butyl propionate from propionic acid and *n*-butanol. These phase equilibrium properties are needed to simulate and design the reactive distillation column for the synthesis of butyl propionate. Recently, Lee et al. [2] studied the kinetics of liquid–solid catalytic esterification of propionic acid with *n*-butanol over Amberlyst 35 at temperatures from 353.15 to 393.15 K. Some phase equilibrium data are available in literature for these related mixtures. Ito and Yoshida [3] measured the VLE data for water + propionic acid at isobaric conditions. Gonzalez and Ortega [4] reported the VLE data for *n*-butanol + butyl propionate, and Stephenson and Stuart [5] measured the mutual solubility data of *n*-butanol + water and butyl propionate + water. While LLE data of the related aqueous systems at ambient conditions are plentiful in literature, the VLLE data are rather limited. In the present study, isothermal VLE data of propionic acid

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+ butyl propionate were measured with a static apparatus over the temperature range of 373.15–393.15 K. The isothermal VLE data were also determined experimentally with the same apparatus for two binary systems of *n*-butanol + water and butyl propionate + water up to 393.15 K and a ternary system of water + *n*-butanol + butyl propionate at 343.15 and 363.15 K. Multiphase equilibrium data for water + *n*-butanol + butyl propionate are not available in literature at these comparable conditions. These new phase equilibrium data are utilized to examine the reliability of the VLE calculations by using the NRTL model [6].

2. Experimental

The static VLE apparatus, installed by Lee et al. [7], was employed in the present study to measure the VLE and the VLE data. The center of the apparatus is the visual equilibrium cell that was immersed in a visibility thermostatic bath (Model TV-4000, stability = ± 0.03 K, Neslab, USA). The phase behavior in the equilibrium cell can be observed through the transparent windows. Bath temperature was measured by a precision thermometer (Model 1506, Hart Scientific, USA) with a platinum RTD probe to an accuracy of ± 0.02 K. A pressure transducer (Model PDCR-912, 0–1000 kPa, Druck, UK) with a digital indicator (Model DPI-261, Druck, UK) measured the equilibrium pressure. The accuracy of the pressure measurement is about $\pm 0.1\%$.

This apparatus was equipped with both liquid and vapor circulation loops to promote the attainment of equilibrium. A four-port liquid-sampling valve (Model 7410, Rheodyne, USA) with a 1 μ l loop disc was installed in the liquid circulation loop to trap the liquid sample, while a pre-heater was used to reheat the returned liquid to the cell if not sampling. A switch valve was utilized to select one of coexistent liquid phases to be circulated. The gas circulation loop consists of a six-port vapor-sampling valve (Model 7010, Rheodyne, USA) with a 100 μ l sample loop, a magnetic gear pump, and a pre-heater.

The composition of sample was analyzed by a gas chromatography (GC) (Model: 8700, China Chromatography Co., Taiwan) with thermal conductivity detector. High purity helium (99.99%) was used as a carrier gas. A stainless steel column packed with 10% Porapak Qs 80/100 (2 m $\times$ 0.3175 cm) can completely separate the constituent compounds of the samples.

A mixture to be studied was loaded in the degassing unit at the beginning of an experiment. The degassing procedure is similar to that of Lee and Hu [8]. The degassed solution was then transferred into the equilibrium cell, in which the levels of the vapor–liquid–liquid interfaces should be adjusted properly such that the upper liquid phase can be circulated. Both liquid and vapor mixtures were circulated alternatively by the circulation pumps to promote equilibration. While the system reaches equilibrium, the pressure reading of the cell approaches a constant. Five samples were replicated for each phase at a fixed experimental condition. The averaged area fraction from GC was converted into mole fraction by the calibration equations. Calibrations were made with gravimetrically prepared samples within two composition ranges, in accordance with those in the organic-rich (including the vapor phase) and the water-rich phases for the ternary system. The uncertainties of the composition analysis for the minor components are estimated to be 0.001 in mole fraction for the organic and the vapor phase and 0.00001 for the aqueous phase.

When the VLE apparatus is operated at lower temperatures, the liquid mixtures may be too viscous to be circulated continuously by the liquid pump that can cause problems of sampling. Under these circumstances, only saturated pressures were measured.

Butyl propionate (>99 mass%), propionic acid (>99 mass%), and *n*-butanol (>99 mass%) were purchased from Aldrich Company, USA. Deionized distilled water was prepared in our laboratory. No impurity peaks were detected by the chromatographic analysis for any one of the chemicals. These chemicals were used without further purification. The physical properties of these compounds are listed in Table 1.

Table 1
Physical properties for the constituent compounds

Component	T_b (K)	T_c (K)	P_c (kPa)	V_c (cm ³ mol ⁻¹)	Z_c	ω	R_g (Å)	μ (D)	C_1	C_2	C_3
<i>n</i> -Butanol	390.9 ^a	563.1 ^a	4420 ^a	275.0 ^a	0.259 ^a	0.593 ^a	3.225 ^b	1.8 ^a	8.5407 ^c	1335.028	−96.496
Propionic acid	414.5 ^a	612.0 ^a	5400 ^a	222.0 ^a	0.183 ^a	0.520 ^a	3.050 ^b	1.5 ^a	8.6746 ^c	1615.23	−68.36
Butyl propionate	419.8 ^d	596.0 ^d	2560 ^d	461.0 ^d	0.238 ^d	0.372 ^d	4.283 ^e	—	6.7566 ^f	1871.62	−26.583
Water	373.15 ^a	647.14 ^a	22064 ^a	55.95 ^a	0.229 ^a	0.344 ^a	0.615 ^c	1.8 ^a	7.11564 ^g	1687.537	−42.98

^a Taken from Reid et al. [9].

^b Taken from Reid et al. [10].

^c Antoine equation: $\log_{10}(P^s) = C_1 - C_2/(T + C_3)$, where P^s in kPa and T in K. The Antoine constants were taken from Poling et al. [11].

^d Taken from the TRC tables [12].

^e Estimated from the method of Harlacher and Braun [13], in which Parachor was calculated by a group contribution method [9].

^f The vapor pressures of butyl propionate were measured in this study and correlated with the Antoine equation as given in footnote c.

^g The Antoine constants were taken from Boublik et al. [14].

3. Experimental results

3.1. Binary VLE of propionic acid + butyl propionate

Fig. 1 presents the pressure–composition diagram for propionic acid + butyl propionate, indicating that its vapor–liquid coexistence region is very narrow and no azeotrope forms in this binary system. The isothermal VLE data are listed in Table 2 for the binary system of propionic acid + butyl propionate at 373.15 and 393.15 K. The tabulated activity coefficients (γ_i) were calculated from the criteria of phase equilibria with the assumption of an ideal vapor-phase, i.e.

$$\gamma_i = \frac{y_i P}{x_i P_i^s \exp[(P - P_i^s)V_i^L/(RT)]} \quad (1)$$

where x_i , y_i , P_i^s and V_i^L are liquid mole fraction, vapor mole fraction, saturated vapor pressure, and liquid molar volume for component i , respectively. In the calculation, the liquid molar volumes were estimated from the modified Rackett model [15] and the saturated vapor pressures were calculated from the Antoine equation with the coefficients tabulated in Table 1. The excess Gibbs free energy (G^E) was calculated from its definition:

$$G^E = RT \sum_{i=1}^{n_c} x_i \ln \gamma_i \quad (2)$$

Table 3 shows the results of the thermodynamic consistency test of the experimental data. All these isothermal VLE data passed the point, the area, and the infinite dilution tests. The method of the consistency test was detailed in Kojima et al. [16].

Table 2

VLE data for propionic acid (1) + butyl propionate (2)

T (K)	P (kPa)	x_1	y_1	$\ln \gamma_1$	$\ln \gamma_2$	$G^E/(RTx_1x_2)$
373.15	22.0	0.0	0.0			
	22.2	0.1151	0.1188	−0.0371	0.0047	−0.0012
	22.4	0.1980	0.2071	−0.0152	0.0063	0.0129
	22.6	0.2702	0.2782	−0.0221	0.0154	0.0268
	22.8	0.3673	0.3733	−0.0262	0.0254	0.0280
	23.0	0.4796	0.4851	−0.0223	0.0328	0.0257
	23.2	0.5830	0.5882	−0.0162	0.0393	0.0285
	23.3	0.6851	0.6956	−0.0057	0.0222	0.0143
	23.5	0.7868	0.7969	−0.0019	0.0139	0.0089
	23.6	0.8971	0.9039	−0.0008	0.0005	−0.0071
	23.8	1.0	1.0			
393.15	45.7	0.0	0.0			
	46.5	0.1178	0.1299	0.0273	0.0029	0.0556
	47.0	0.1995	0.2103	−0.0076	0.0135	0.0581
	47.5	0.2773	0.2922	0.0020	0.0166	0.0625
	48.0	0.3659	0.3802	−0.0014	0.0245	0.0648
	48.6	0.4709	0.4981	0.0284	0.0066	0.0678
	49.1	0.5852	0.6003	0.0077	0.0321	0.0735
	49.3	0.6864	0.6933	−0.0038	0.0509	0.0620
	49.6	0.7838	0.8002	0.0128	0.0001	0.0595
	49.8	0.8961	0.9007	0.0012	0.0376	0.0534
	50.0	1.0	1.0			

3.2. Binary VLLE of *n*-butanol + water and butyl propionate + water

Table 4 lists the equilibrium pressures and the phase compositions for the binary systems of *n*-butanol + water and butyl propionate + water at VLLE in a temperature

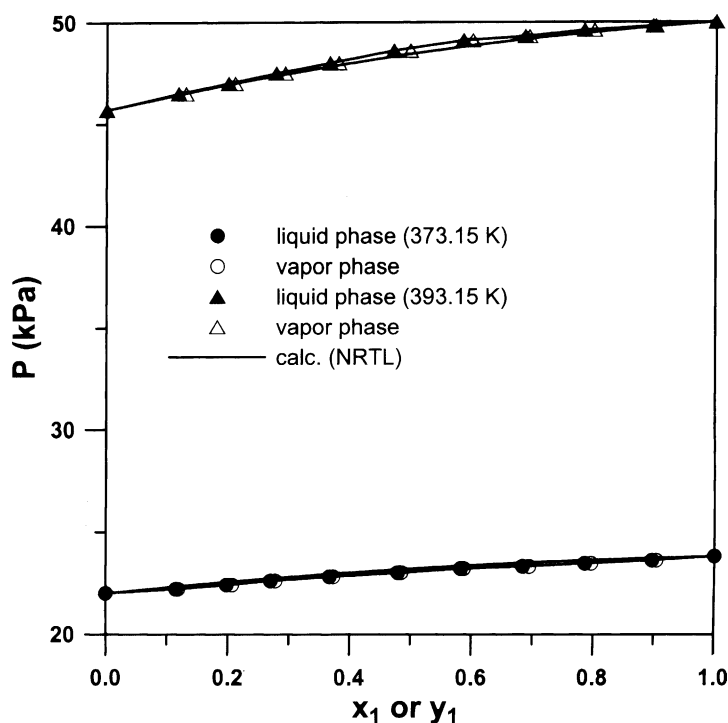


Fig. 1. Pressure–composition diagram for propionic acid (1) + butyl propionate (2).

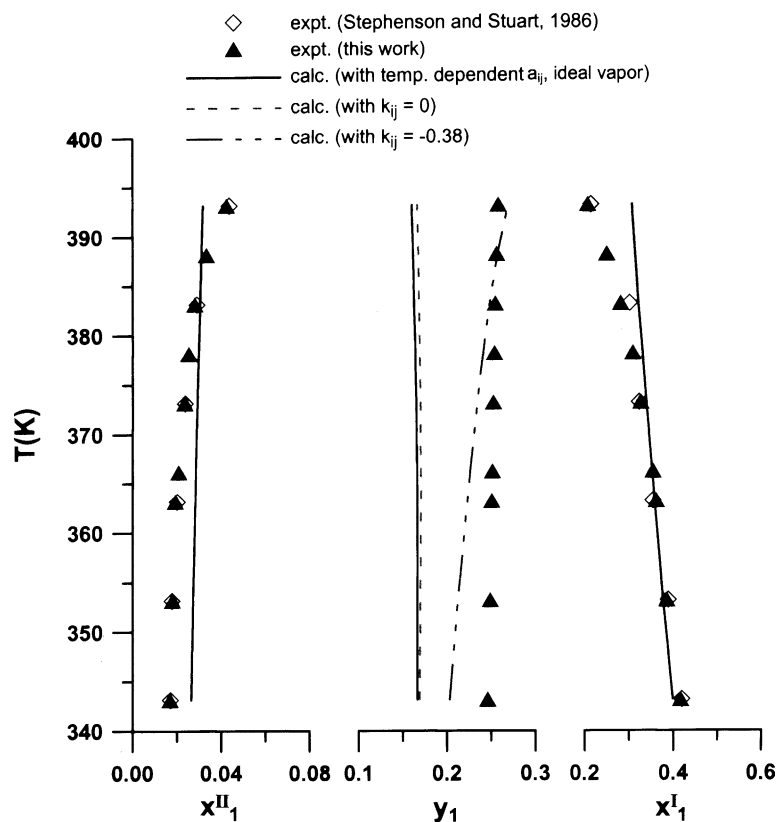


Fig. 2. Temperature–composition VLE phase diagram for *n*-butanol (1) + water (2).

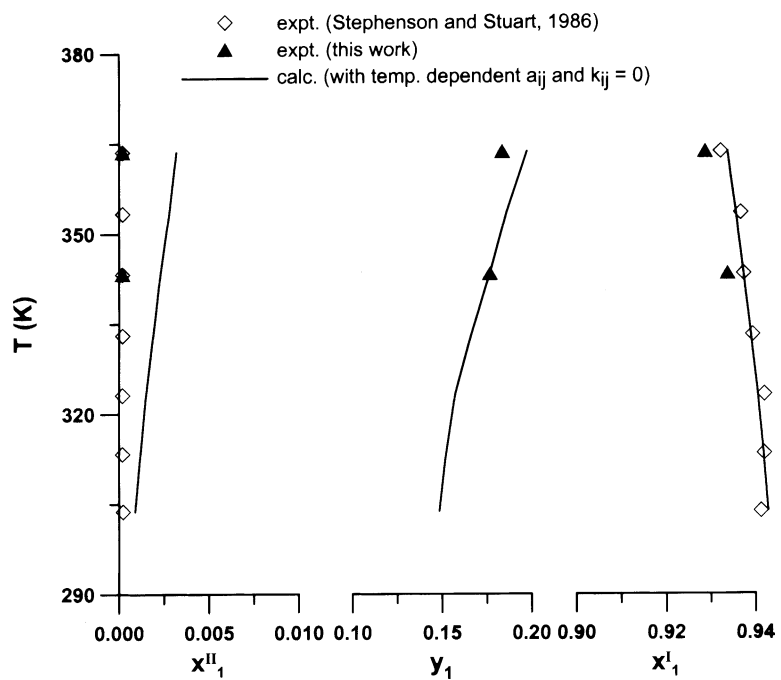


Fig. 3. Temperature–composition VLE phase diagram for butyl propionate (1) + water (2).

Table 3
Results of the thermodynamic consistency test for propionic acid + butyl propionate

T (K)	Consistency test index ^{a,b}				G^E/RT coefficient ^c			$\ln(\gamma_1/\gamma_2)$ coefficient ^d			
	δ	A	I_1	I_2	C_0	C_1	C_2	D_0	D_1	D_2	D_3
373.15	3.92 (+)	3.00 (+)	27.43 (+)	25.26 (+)	0.0288	−0.0058	−0.0524	−0.0299	−0.0183	−0.0374	−0.0137
393.15	2.89 (+)	1.19 (+)	16.42 (+)	18.28 (+)	0.0683	0.0001	−0.0245	−0.0119	0.0221	−0.0112	0.0308

^a Criteria for passing the thermodynamic consistency tests: $\delta < 5$, $A < 3$, $I_1 < 30$, $I_2 < 30$.

^b (+): Passes the consistency test.

^c $G^E/RT = x_1x_2[C_0 + C_1(x_1 - x_2) + C_2(x_1 - x_2)^2]$.

^d $\ln(\gamma_1/\gamma_2) = D_0 + D_1(x_2 - x_1) + D_2(6x_1x_2 - 1) + D_3(x_2 - x_1)(1 - 8x_1x_2)$.

Table 4
Binary vapor–liquid–liquid equilibrium data

T (K)	P (kPa)	Organic phase, x_1^I	Aqueous phase, x_1^{II}	Vapor phase, y_1
<i>n</i> -Butanol (1) + water (2)				
323.15	14.4	–	–	–
333.15	24.4	–	–	–
343.15	39.6	0.4165	0.01678	0.2460
353.15	61.5	0.3857	0.01798	0.2486
363.15	92.5	0.3610	0.01924	0.2500
366.15	102.4	0.3535	0.02077	0.2507
373.15	134.4	0.3257	0.02347	0.2517
378.15	160.6	0.3078	0.02525	0.2531
383.15	191.3	0.2808	0.02796	0.2539
388.15	226.6	0.2493	0.03313	0.2552
393.15	267.0	0.2059	0.04215	0.2570
Butyl propionate (1) + water (2)				
343.15	38.2	0.9335	0.00019	0.1765
363.15	86.5	0.9285	0.00019	0.1831

range of 323.15–393.15 K. Figs. 2 and 3 illustrate compositions of the coexistence phases of the organic constituent varying with temperature for *n*-butanol + water and butyl propionate + water, respectively. The literature values of the mutual solubilities of these two aqueous systems [5] are also shown in the graphs for comparison. In general, the agreement is reasonably well. Fig. 4 is the P – T diagram at VLLE for *n*-butanol + water. The saturated pressures varying with temperature appear to follow the Antoine relationship.

3.3. Ternary VLLE of water + *n*-butanol + butyl propionate

The VLLE data were measured for the ternary system of water + *n*-butanol + butyl propionate at 343.15 and 363.15 K. The experimental results are reported in Table 5, in which the superscript I represents the organic-rich phase

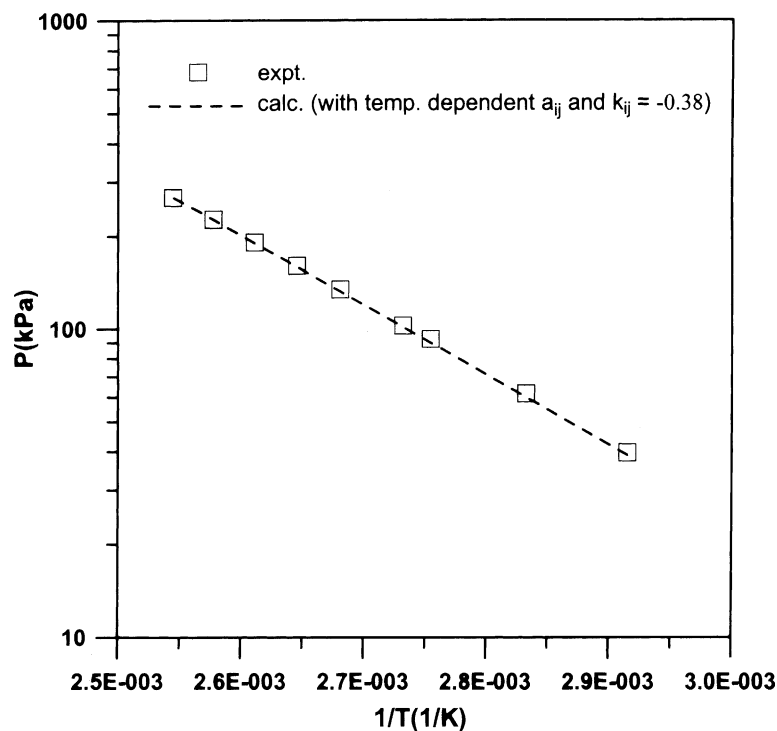


Fig. 4. Saturated pressures at VLLE for *n*-butanol + water.

Table 5
VLE data for water + *n*-butanol + butyl propionate

<i>T</i> (K)	<i>P</i> (kPa)	Organic phase, x_i^I		Aqueous phase, x_i^{II}		Vapor phase, y_i	
		<i>n</i> -Butanol	Butyl propionate	<i>n</i> -Butanol	Butyl propionate	<i>n</i> -Butanol	Butyl propionate
343.15	39.6	0.4165	–	0.01678	–	0.2300	–
	39.3	0.4494	0.0947	0.01495	0.00003	0.1494	0.0566
	38.9	0.3836	0.2926	0.01060	0.00006	0.0993	0.1249
	38.6	0.2485	0.5381	0.00813	0.00012	0.0687	0.1378
	38.4	0.1611	0.6850	0.00646	0.00016	0.0676	0.1442
	38.2	–	0.9335	–	0.00019	–	0.1765
363.15	92.5	0.3610	–	0.01924	–	0.2300	–
	91.1	0.4140	0.0973	0.01535	0.00004	0.1678	0.0465
	90.0	0.3581	0.3001	0.01046	0.00005	0.1279	0.0886
	88.8	0.2527	0.5178	0.00789	0.00015	0.0794	0.1266
	87.9	0.1635	0.6471	0.00551	0.00017	0.0624	0.1364
	86.5	–	0.9285	–	0.00019	–	0.1831

and II, the aqueous phase. Figs. 5 and 6 are the phase diagrams for this ternary system at 343.15 and 363.15 K, respectively. Among these three constituent compounds, *n*-butanol and butyl propionate are completely miscible. Both *n*-butanol and butyl propionate, however, are partially miscible with water, resulting that this ternary system exhibits the type 2 LLE behavior as shown in the figures. The solubilities of the butyl propionate are rather small in the aqueous phase (<0.02%), while water dissolves appreciably in the organic-rich phase, especially in the region where *n*-butanol is the major compound. Meanwhile, the curve of saturated vapor is located in between two coexistence liquid phases. Since the mutual solubilities increase with increasing temperature, the area of liquid–liquid splitting region becomes smaller at higher temperatures.

4. VLE calculation

The binary VLE data of propionic acid + butyl propionate were correlated with ϕ – γ method. Because the equilibrium pressures are sufficiently low, the vapor phase is assumed as ideal mixtures. The optimal values of the NRTL parameters were determined on the basis of the maximum likelihood principle by minimization of the following objective function π :

$$\pi = \sum_{k=1}^{n_p} \left\{ \left[\frac{(P_k^{\text{calc}} - P_k^{\text{expt}})}{\sigma_p} \right]^2 + \left[\frac{(T_k^{\text{calc}} - T_k^{\text{expt}})}{\sigma_T} \right]^2 + \left[\frac{(x_{1,k}^{\text{calc}} - x_{1,k}^{\text{expt}})}{\sigma_{x_1}} \right]^2 + \left[\frac{(y_{1,k}^{\text{calc}} - y_{1,k}^{\text{expt}})}{\sigma_{y_1}} \right]^2 \right\} \quad (3)$$

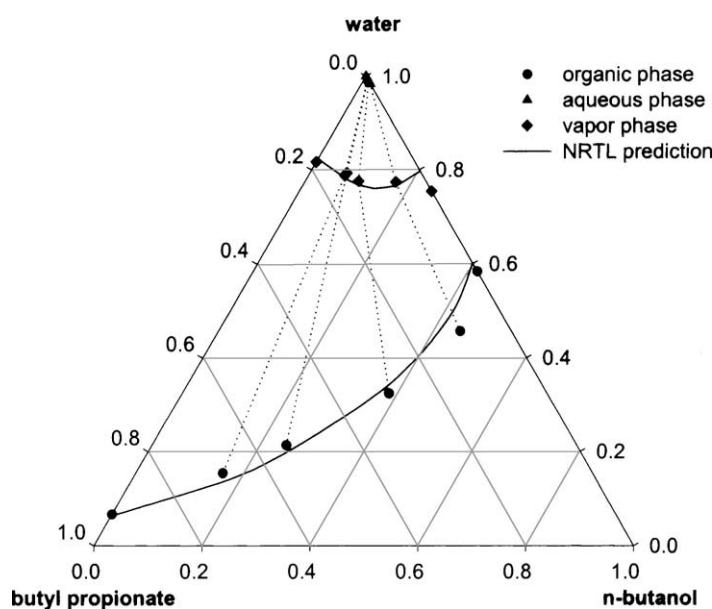


Fig. 5. VLE phase diagram for water (1) + *n*-butanol (2) + butyl propionate (3) at 343.15 K.

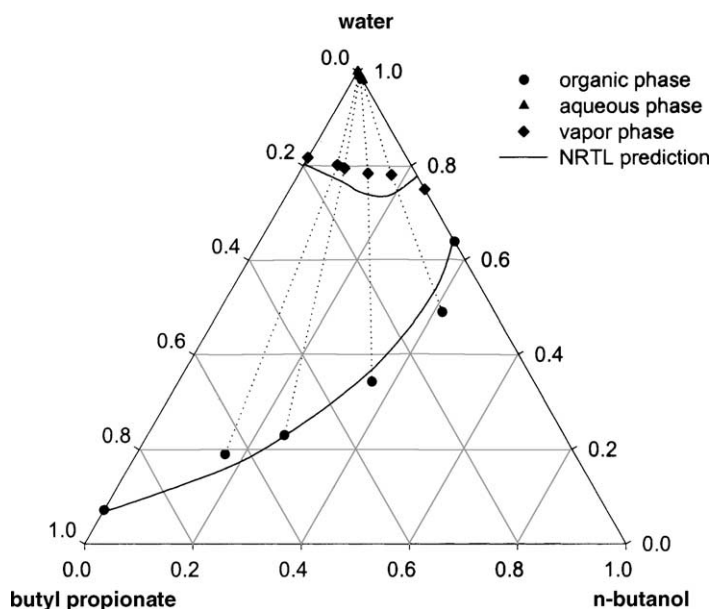


Fig. 6. VLLE phase diagram for water (1) + *n*-butanol (2) + butyl propionate (3) at 363.15 K.

Table 6

The optimized parameters of the NRTL model determined from binary phase equilibria data

<i>i</i> – <i>j</i>	<i>n</i> -Butanol + water	Butyl propionate + water	<i>n</i> -Butanol + butyl propionate	Propionic acid + butyl propionate
$a_{ij,0}^a$	998.73	271.74	–108.03	–133.96
$a_{ji,1}^a$	–4.2649	0.3983	0.0	0.0
$a_{ji,1}^b$	–900.65	2107.5	315.78	174.82
$a_{ji,1}^b$	7.8294	–1.0017	0.0	0.0
α	0.2	0.2	0.3	0.3
k_{ij}	–0.38	0.0	$B_{ij} = 0$	$B_{ij} = 0$
Data type	VLLE	VLLE	VLE ^c	VLE

^a $a_{ij}(K) = a_{ij,0} + a_{ij,1}T(K)$.

^b $a_{ji}(K) = a_{ji,0} + a_{ji,1}T(K)$.

^c Taken from ref. [4].

The standard deviations σ used in the calculation were 0.05 kPa for pressure, 0.05 K for temperature, 0.002 for liquid composition, and 0.005 for vapor composition, respectively. The optimization algorithm is similar to that of Prausnitz et al. [17]. Table 6 reports the optimized parameters of the NRTL model. The smooth curves in Fig. 1 are the calculated results from the NRTL model.

5. VLLE calculation

According to the phase rule, the degree of freedom is unity for a binary system at VLLE. Provided that temperature is given, the equilibrium pressure and any other intensive properties in individual phases of the system are all specified. Based on 1 mol of feed with total composition of z_i (component *i*), the compositions of the coexistent liquid phases (x_i^I and x_i^{II}) at a given temperature can be solved numerically from the criteria of liquid–liquid equilibrium (equality of the constituent fugacities between the coexistence liquid

phases) together with the material balance equation, simultaneously, i.e.

$$z_i = (1 - \beta)x_i^{II} + \beta x_i^I \quad (4)$$

or

$$1 - \sum_{i=1}^{n_c} \frac{z_i}{\beta + K_i(1 - \beta)} = 0 \quad (5)$$

with

$$K_i = \frac{x_i^{II}}{x_i^I} = \frac{\gamma_i^I}{\gamma_i^{II}} \quad (6)$$

where β is the fraction of the total material that is present in the first liquid-phase (i.e. the organic-rich phase, denoted as superscript I), n_c the number of components, the superscript II represents the aqueous phase, and K_i the distribution ratio for component *i*. The activity coefficient γ_i was calculated from the NRTL model in this study. The calculation procedure was detailed in Walas [18].

Since the equilibrium pressures and the vapor pressures of the constituent compounds are not significantly high over the investigated conditions, the Poynting pressure correction factor was negligible and the fugacity coefficient of vapor phase approached to unity in the calculation. The vapor composition (y_i) and the bubble-point pressure (P) can thus be calculated from the following equations:

$$y_i = \frac{\gamma_i^I x_i^I P_i^S}{\phi_i P} = \frac{\gamma_i^{II} x_i^{II} P_i^S}{\phi_i P} \quad (7)$$

and

$$P = \sum_{i=1}^{n_c} y_i P = \sum_{i=1}^{n_c} \frac{\gamma_i^I x_i^I P_i^S}{\phi_i} = \sum_{i=1}^{n_c} \frac{\gamma_i^{II} x_i^{II} P_i^S}{\phi_i} \quad (8)$$

where P_i^S and ϕ_i are the vapor pressure and fugacity coefficient of component i , respectively. The Antoine constants for each compound are listed in Table 1. Provided that vapor phase is assumed as an ideal gas mixture, the fugacity coefficient ϕ_i is unity. Otherwise, the non-ideality of vapor phase was represented by the two-term virial equation, i.e.

$$\ln \phi_i = \frac{2}{V} \sum_{j=1}^{n_c} y_j B_{ij} - \ln Z \quad (9)$$

The method of Lee and Chen [19] was applied to estimate the second virial coefficients. In the VLLE calculations for a binary system, there are two binary parameters (a_{12} and a_{21} with $\alpha = 0.2$) in the NRTL model and one binary interaction parameter (k_{ij}) in the combining rule of T_{cij} for calculation of cross second virial coefficient. These model parameters were determined from the phase equilibrium data of the constituent binaries, and then used in the prediction of VLLE behavior for the ternary system.

5.1. Binary system

For the system of n -butanol (1) + water (2), the temperature-specific parameters of the NRTL (a_{12} and a_{21} with $\alpha = 0.2$) were determined from the mutual solubility data reported by Stephenson and Stuart [5]. These two parameters appear to be a linear function of temperature

over a temperature range of 298.15–393.15 K. The linear relationships are given by

$$a_{12}(\text{K}) = 998.73 - 4.2649T(\text{K}) \quad (10)$$

$$a_{21}(\text{K}) = -900.65 + 7.8294T(\text{K}) \quad (11)$$

With these temperature-dependent parameters, the mutual solubilities can be calculated satisfactorily, except at temperatures higher than 383.15 K, as shown in Fig. 2.

The mole fractions of n -butanol in the vapor phase are underestimated, if the vapor phase is assumed as an ideal gas mixture. As seen from Fig. 2, substantial improvement was achieved by using the two-term virial equation with $k_{ij} = -0.38$ to calculate the fugacity coefficients ϕ_i . The optimized parameters of the NRTL model and the calculated results are tabulated in Tables 6 and 7, respectively.

Based on the LLE data of butyl propionate (1) + water (2) from Stephenson and Stuart [5], the NRTL parameters a_{12} and a_{21} were also found to be linearly depended on temperature. These two temperature-dependent functions are given by

$$a_{12}(\text{K}) = 271.74 + 0.3983T(\text{K}) \quad (12)$$

$$a_{21}(\text{K}) = 2107.5 - 1.0017T(\text{K}) \quad (13)$$

Fig. 3 compares the calculated results with the experimental values, revealing that the NRTL model correlates well the compositions of organic-rich phase but overestimates those of aqueous phase. Since the calculated vapor compositions are relatively insensitive to the value of k_{ij} , the second virial coefficients were estimated with zero k_{ij} . The calculated vapor compositions agree fairly well with the experimental results. The optimized parameters of the NRTL model are reported in Table 6.

5.2. Ternary system

With a similar method as mentioned above, the VLLE properties were predicted for the ternary system of water (1) + n -butanol (2) + butyl propionate (3) by using the model parameters determined from the phase equilibrium data of the constituent binaries. Table 6 gives the NRTL parameters and the k_{ij} values for the constituent pairs.

Table 7
Calculated results of n -butanol (1) + water (2)

Treatments	$\Delta P/P$ AAD ^a (%)	Organic phase, Δx_1^I AAD ^b	Aqueous phase, Δx_1^{II} AAD ^b	Vapor phase, Δy_1 AAD ^b
Temperature-dependent a_{ij} ^c ideal vapor phase	12.5	0.0301	0.0069	0.0880
Temperature-dependent a_{ij} ^c virial equation with $k_{ij} = 0.0$ ^d	11.8	0.0299	0.0069	0.0811
Temperature-dependent a_{ij} ^c virial equation with $k_{ij} = -0.38$ ^d	1.2	0.0299	0.0069	0.0206

^a $\Delta P/P$ AAD (%) = $(100/n_p) \sum_{i=1}^{n_p} (|P^{\text{calc}} - P^{\text{expt}}|/P^{\text{expt}})_i$, where n_p is the number of data points.

^b ΔM AAD = $(1/n_p) \sum_{i=1}^{n_p} (|M^{\text{calc}} - M^{\text{expt}}|)_i$, where n_p is the number of data points and ΔM represents Δx_1^I , Δx_1^{II} , or Δy_1 .

^c $a_{12}(\text{K}) = 998.73 - 4.2649T(\text{K})$, $a_{21}(\text{K}) = -900.65 + 7.8294T(\text{K})$, and $\alpha_{12} = 0.2$.

^d B_{ij} was calculated from the method of Lee and Chen [19].

Table 8
Predicted results of VLLE for water (1) + *n*-butanol (2) + butyl propionate (3)

Grand AAD ^a	$T = 343.15 \text{ K}$ ($\Delta P/P = 1.34\%$)	$T = 363.15 \text{ K}$ ($\Delta P/P = 2.72\%$)
Organic phase		
Δx_1^I	0.0247	0.0277
Δx_2^I	0.0137	0.0143
Δx_3^I	0.0110	0.0135
Aqueous phase		
Δx_1^{II}	0.0045	0.0065
Δx_2^{II}	0.0026	0.0037
Δx_3^{II}	0.0019	0.0028
Vapor phase		
Δy_1	0.0197	0.0434
Δy_2	0.0080	0.0300
Δy_3	0.0236	0.0100

^a $\Delta P/P$ Grand AAD (%) = $(100/n_p) \sum_{i=1}^{n_p} (|P^{\text{calc}} - P^{\text{expt}}|/P^{\text{expt}})_i$, where n_p is the number of data points. ΔM Grand AAD = $\sum_{i=1}^{n_{\text{TL}}} \sum_{j=1}^{n_c} (|M_{i,j}^{\text{calc}} - M_{i,j}^{\text{expt}}|)_i / (n_c \times n_{\text{TL}})$, where n_c is the number of components, n_{TL} the number of tie-lines, i the index of tie-line, j the index of component, and ΔM represents Δx^I , Δx^{II} , or Δy .

While the parameters of *n*-butanol/water and butyl propionate/water at a specific temperature were obtained from their mutual solubility (LLE) data (i.e. Eqs. 10–13), those of *n*-butanol/butyl propionate were determined from binary VLE data [4]. The predicted results are reported in Table 8. Figs. 5 and 6 compare the predictions with the experimental values for the ternary system at 343.15 and 363.15 K, respectively. For a type 2 of LLE system, the key to successfully predict the phase boundaries is that the model should reproduce equally well the mutual solubilities of the two partially miscible binary systems over the entire temperature range. These two graphs show that the NRTL model with two-term virial equation predicts the boundaries of organic-rich phase accurately and the vapor compositions reasonably.

6. Conclusion

Isothermal VLLE properties were determined experimentally for two binary and one ternary aqueous systems containing *n*-butanol and butyl propionate at temperatures up to 393.15 K. The ternary system behaves as a type 2 of LLE. The VLE data were also measured for propionic acid + butyl propionate at 373.15 and 393.15 K. No azeotrope formed in this system. The NRTL model with temperature-independent energy parameters is inadequate to represent accurately the mutual solubilities for these two binary aqueous systems at temperatures up to 393.15 K. Significant improvement has been obtained by assuming that these parameters were linearly dependent on temperature. Estimating the non-ideality of the vapor phase from the second virial equation yielded better results for the saturated vapor compositions for water

+ *n*-butanol. By using the parameters determined from the constituent binaries, the NRTL model with the two-term virial equation is capable of reproducing the VLLE properties for the ternary system of water + *n*-butanol + butyl propionate.

List of symbols

a_{ij}	parameters of the NRTL model ($g_{ij} - g_{ji}$)/ R (K)
A	index of area consistency test
C_1, C_2, C_3	Antoine constants
G	Gibbs free energy (J mol^{-1})
I_1, I_2	indices of infinite dilution consistency test
k_{ij}	interaction constant for calculation of cross virial coefficient B_{ij}
K_i	distribution ratio for component i
MW	molecular weight (g mol^{-1})
n_c	number of components
n_p	number of data points
n_{TL}	number of tie lines
P	pressure (kPa)
R	gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
R_g	radius of gyration ($\text{\AA}$)
T	temperature (K)
V	molar volume ($\text{cm}^3 \text{mol}^{-1}$)
x	mole fraction in liquid phase
y	mole fraction in vapor phase
z_i	mole fraction of component i in feed
Z	compressibility factor

Greek letters

α	parameter of the NRTL model
β	fraction of the total material in the organic-rich phase
γ	activity coefficient
δ	index of point consistency test
μ	dipole moment (D)
π	objective functions
σ	standard deviation
ϕ_i	fugacity coefficient for component i

Subscripts

b	boiling
c	critical
i, j	components i and j
ij	i – j pair interaction

Superscripts

calc	calculated
expt	experimental
E	excess property
L	liquid phase
s	saturation
I	organic phase
II	aqueous phase

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